

CBM+ Industry Day Questions

When you mention “Condition Based maintenance Plus (CBM+), is it the same as “Predictive Maintenance”, since the RFI indicates the use of AI/ML?

Answer: Correct. CBM+ and Predictive Maintenance in this instance are synonymous

Is this a new requirement or recompetete and if the latter; who is the current incumbent?

Answer: This is the first time CBM+ is being competed. We are currently using DTIC IAMAC contract. KBR is the incumbent.

What is the rough order of magnitude (ROM), if known?

Answer: We are not able to discuss funding at this point.

Will this potential opportunity be an IDIQ multiple award contract (MAC)?

Answer: We are seeking an IDIQ FFP contract, but not a multiple award.

What is the period of performance (PoP), and maximum funding (if known)?

Answer: 3 years with 2 option years. We are not prepared to discuss funding at this point.

Do you anticipate a small business set aside for this effort or part of it?

Answer: To be Determined. This has not been reviewed by the small business office yet.

Do you anticipate the vendors’ to have cleared personnel at the time of proposal submission?

Answer: Yes

Do you envision the adaptation of XR to be included to this effort now or in the future?

Answer: If XR is to be included in the CBM+ capability it will be a future effort.

How long has the current system and associated dashboard been running?

Answer: Since 2023

Who manages this current system, and is that entity the one who built the environment?

Answer: NSWC Crane and KBR

As this is a new contract (no incumbent) just looking to frame the current landscape

Answer: While this is an open competition contract, we built the current system using a DTIC contract. KBR is the current incumbent.

Is this purely for land machinery/vehicles or does it span to sea/air assets?

Answer: Land only

Is there a POC phase on a limited scale of assets?

Answer: A proof of concept is up and running now

Do contract requirements include bringing on experts and conducting pain point analysis to what types of sensors are most needed or is this already determined?

Answer: CBM+ relies on the platform program offices to conduct RCM analysis.

I see "oil life" listed, which can be difficult environment for new sensors, is there a notion of how many sensor types are needed or if some will come from interfacing with existing machine readings? (Example sensor types: pressure/depth, vibration, heat, electrical (various), humidity, level / angular, pressure, volume, rotations/usage, altitude, IR, distance etc.)

Answer: CBM+ is not installing additional sensors on platforms. Oil life is being calculated based on information available on the CANBus

Is the intention for static/installed sensors or for mobile sensors that could either aid in troubleshooting or serve a larger fleet?

Answer: CBM+ installs static equipment that records, stores, and transmits data available on the CANBus. No additional sensors are installed.

Is there a clean source of record that records asset failures?

Answer: Failure data is available in the Global Combat Support System-MC (GCSS-MC)

Is the data all needing to sync to a centralized data store?

Answer: Data is stored on the Government provided cloud environments.

Does the data store also need to be locally managed (e.g. units retrieving asset history) for when unit may have limited connection?

Answer: No

Does the system need to sync with existing maintenance CMMS, supply systems or other systems of record?

Answer: Yes. But it will take time and coordination to establish these connections

Assuming connectivity, what is the expected time from warning to insight?

Answer: 8 hours

Any specific things that we are not doing now that we would like to see in a new contract.

Answer: I'm not able to answer this question.

What is the anticipated timeline for the release of the formal RFP, including any draft RFP periods, and will there be opportunities for one-on-one meetings or site visits following Industry Day?

Answer: The program office does not intend to conduct one-on-one meetings after this industry day. I'll defer to the contract office on the timeline.

Can you provide more details on the specific ground vehicle platforms that the CBM+ hardware must integrate with, including any variations in CANbus configurations across different Marine Corps vehicles?

Answer: Medium Tactical Vehicle Replacement (MTVR) and Logistics Vehicle System Replacement (LVSR) CANBus 1708 and 1939 protocols. Joint Light Tactical Vehicle (4 Busses) CANBus 1939 protocol. Medium Crawler Tractor (MCT) CANBus 1939 protocol. Hydraulic Excavator (HYEX) CANBus 1939 protocol. Tractor, Rubber-Tired, Articulated Medium (TRAM) CANBus 1939 protocol. Medium Tractor Loader (MTL) CANBus 1939 protocol. Ultra-Light Tactical Vehicle (ULTV) CANBus 1939 protocol. Amphibious Combat Vehicle (ACV) (4 Busses) CANBus 1939 protocol. Remotely Operated Ground

Unit for Expeditionary Fires (ROUGE) CANBus 1939 protocol and a Robotics bus that is yet to be evaluated. Other platforms as required.

Regarding the hardware size constraint (no larger than 7.5" x 6" x 3" including mounting), are there any flexibility allowances for ruggedization or environmental protection features?

Answer: No

Will the government provide detailed interface control documents (ICDs) or APIs for integrating with GCSS-MC, EFAS, and other existing systems prior to the solicitation release?

Answer: Yes

In the cloud storage requirements, access is mentioned for Microsoft Azure on MCEN and Navy's Jupiter in Advana, will contractors receive government-furnished accounts or credentials for development and testing, and what level of classification apply

Answer: Yes. Right now, classification is CUI but should plan for SECRET.

For AI/ML analysis, what specific performance metrics (e.g., accuracy thresholds for failure prediction or remaining useful life estimation) will be used to evaluate models, and are there baseline datasets available for benchmarking?

Answer: The program is not at the point of being able to predict a part failure or remaining useful life due to the absence of performance metrics for these parts. Our intent is to use failure data in GCSS-MC to develop this capability.

Regarding data rights, the government retains all rights to models, source code, and visualizations, will proposals need to include unlimited rights data (URD) clauses, and are there restrictions on using open-source or commercial off-the-shelf (COTS) components?

Answer: Yes. Restrictions on using open source or COTS components exists due to cyber security levels on government cloud environments. Approval of open-source and COTS components will be on a case-by-case basis.

In the PWS, MIL-STD-882 for system safety is mandatory, can you provide guidance on the expected safety hazard analysis level (e.g., preliminary vs. full) and integration with RMF (DoD 8510.01)?

Answer: The Government will conduct a full hazard analysis with the contractor in a supporting role. The RMF process will be adhered to. Cyber Security work will be required.

The PWS specifies delivery of up to 3,000 data loggers, can you clarify the anticipated ordering periods, quantities per period, and any options for surge capacity?

Answer: The projected ordering quantities are TBD and based on funding. Should funding be available surge capacity should be available.

For provisioning requirements, including the Provisioning Parts List (PPL) and Engineering Data for Provisioning (EDFP), will the government facilitate access to the IUID Registry and Defense Logistics Agency systems for pre-screening and registration?

Answer: The Government will assist as required.

For the delivery or update of system documentation, IAW CDRL A006 Software Design Description, which of the listed bullet items already have a baseline or draft version?

Answer: All of them.

For the FARS additional data source for fuel quantities, locations, and movement is the data visualization in the separate and distinct dashboard in the Jupiter environment in scope of the solicitation tasking, or just the data interface and integration?

Answer: FARS will have a separate Dashboard.

For the EMSS additional data source, if the offeror recommends a solution with additional at-platform capability to provide resiliency and data integrity for transfer, store, analyze, and act within a denied, disrupted, intermittent, and limited bandwidth environment, will EMSS capability additions be considered within the scoring criteria for value add.

Answer: All tactical edge solutions will be evaluated for their value

The PWS specifies that the Data Logger Hardware must meet all Performance Specification Requirements. When will the performance specification be made available?

Answer: The P-SPEC will be available with the RFP release

Is it anticipated that the solicitation scoring criteria will accommodate equivalent work to that described in the PWS performed as a subcontractor as well as a prime contractor?

Answer: All recent and relevant work will be evaluated.

Is it anticipated that the solicitation scoring criteria will accommodate the combined capabilities of a team with equivalent work to that of the PWS, including experience of both a prime offeror and its subcontractor team members

Answer: Yes

[Scope] What is the government's definition of "CBM+ capability delivered" at acceptance minimum viable end state across hardware, pipeline, models, dashboards, training, and ATO?

Answer: "CBM+ capability delivered" is not a final, static product, but rather the functioning framework that enables a data-driven, predictive approach to maintenance. This framework must provide a verifiable potential to increase materiel availability and reduce total ownership costs, with the full intent of providing the opportunity for continuous refinement and improvement throughout the equipment's lifecycle.

[Success Measures] What are the top operational outcomes for Year 1 (readiness impacts, reduced NMC time, fewer unscheduled removals), and how will success be measured?

Answer: Readiness impacts, reduced NMC time, fewer unscheduled removals, and reduction in maintenance/operational costs are the top operational outcomes for Year 1. Success in readiness impacts and reduced NMC time may include measurements in Mission Capable Rate and in Non-Mission Capable Maintenance and Non-Mission Capable Supply time. Success in fewer unscheduled removals may include measurements in the number of unscheduled maintenance actions and the ratio of planned vs unscheduled maintenance. Success in reduction in maintenance/operational costs may include measurements in maintenance cost per operating hour.

[Acceptance Criteria] Will the government define acceptance criteria per major capability area (hardware installs, data pipeline, dashboards, models, integrations), or should the contractor propose them?

Answer: Yes

[Lab Requirements, PWS 3.3.4] What constitutes an acceptable "CBM+ lab environment" (required hardware configs, vehicle simulators, test data sets), and does the government have location or access expectations for this lab?

Answer: The current lab is established at NSWC Crane and consists of data logger hardware, hardware configurations, CANBus Emulators, a representative vehicle, and a wireless connection to the MCEN. The contractor can also establish an on-prem lab for their internal use for testing.

[Spare Parts, PWS 4.3.16] When will the USMC maintenance strategy decision be made, and how will it drive spare part quantities and the repair concept under this contract?

Answer: Unknown. Policy documents are being revised at DC I&L

Section 2.1: Marine Corps Order (MCO) 4151.22 Condition Based Maintenance Plus (CBM+) Order directs the adoption of USMC CBM+; what has been the adoption rate thus far? What specific projects have been executed and/or what metrics have been reported, to date? Related, what is the greatest issue holding back CBM+ adoption at full scale?

Answer: We have 14 CBM+ enabled units across all III MEFs, however actual usage/adoption is unit dependent. Some are very good adopters and some are poor adopters. When CBM+ is adopted fully we have seen maintenance turnaround times reduced by 40 days or more. Full scale adoption is hindered by fielding levels, policy, and training.

Section 3.1.4: would “legacy system **integration or** retirement” be acceptable? If no, why not?

Answer: The intent is to retire the legacy system when the V2 system comes online.

Sections 3.2.2 and 4.3.2 – Can the Post Award Conference and Provisioning Conference be held virtually or at a rented facility?

Answer: Both are acceptable.

Section 3.3.1: Will access to Global Combat Service Support-Marine Corps (GCSS-MC) be provided, too?

Answer: CBM+ gets daily data feeds from GCSS-MC. To date, access to the actual system has not been required. Changes to GCSS are controlled by PEO MLB and we work with them to modify the system.

Section 3.3.4: Does “bumper number” = serial number?

Answer: Yes.

Section 3.3.4 states: “Perform cyber security scans and Security Technical Implementation Guides (STIG) reviews in support of annual security reviews and

Authority to Operate (ATO) attainment/sustainment”. Is there a schedule or process defining permission requests and grants?

Answer: Yes. The Risk Management Framework is a well-defined process. Access to systems to perform these functions will be granted to qualified personnel.

Section 3.3.9: Is there a reason the “AN/TPS-80 Ground/Air Task-Oriented Radar (G/ATOR)” is not listed? Can details be shared as to why?

Answer: This program is currently focused on rolling stock with CANBus systems that can be recorded. Branching out to equipment like G/ATOR would be a future effort.

Section 3.4.1: Is a full-time on-site placement expected for Camp Pendelton, Camp Lejeune and Okinawa or is having a person on-site just during install and regular check ins is sufficient?

Answer: A full time FSR at each location is expected.

Section 3.4.2 mentions Hawaii but has no base name. Is it correct to assume MCAS Kaneohe Bay?

Answer: Yes

Section 3.6: Can access to additional systems be requested?

Answer: Please clarify question. 3.6 speaks to stakeholder management not systems.

Section 3.12: Does this section on Information Assurance assume the contractor will define the authoritative requirements for compliance or will this requirements/guidance be provided by the USMC?

Answer: The RMF requirements/guidance will be provide by the USMC

Section 5.0.1 – Will travel included in our proposal be considered COR approved at time of contract award or will proposed travel need approved by COR during contract execution (on a trip by trip basis)?

Answer: Travel will be approved by the COR on a trip by trip basis.

Is there an incumbent?

Answer: the program is on a DTIC contract through NSWC Crane with KBR as the incumbent Prime.

Will the effort be issued as a multi-year program or a single period of performance?
Also, will the award be issued as a T&M or FFP contract type?

Answer: We are seeking a 3 year plus 2 option years FFP contract.

What are the top three metrics that must be reported to demonstrate success?

Answer: Decreased unscheduled maintenance, increased operational availability, and Increased readiness.

Will USMC personnel be available for validation confirmation?

Answer: Yes

Is the CAN Bus operational on all vehicles or does the hardware need to be able to support a different type of Bus such as a 1553?

Answer: This program is concentrated on rolling stock. 1553 is typically used for aviation.

Will Application Program Interfaces (APIs) be available to Azure and Advana?

Answer: Yes

In accordance with CDRL A005, what format does the code need to be in?

Answer: Code is stored in Fusion Spork+ and GitLab. Access to these repositories will be provided.

The Industry Day document states “The Government shall retain all rights to models, source code, and visualizations created in the provided cloud environment. All data and information input into the solution by the customer remains the sole property of the customer.” This language contradicts FARs/DFARs intellectual property clauses which state that the Government has rights, but title and ownership are retained by the contractor. Please clarify the data rights statements are not regarding ownership. Also, related, what is the USMC position on using proprietary software platforms? Or is the desire to not license any portion of the solution?

Answer: The desire is not to license any portion of the solution. The intent of “owning the information/solution” is that the USMC is not paying licensing or subscription fees for models and other software created and the solution remains functional should a recompetes result in a different vendor.

Does data collection come from only existing sources, such as thru the CAN buss, or do you anticipate the need for addition data collection thru deployment of additional sensing capability to augment existing data collection where applicable.

Answer: Adding additional sensors to the platform would be the Platform Owners responsibility based on their RCM analysis.

What are the current SWAP-C requirements (we see the footprint of 7.5"x 6" x 3") but what about the WAP-C?

Answer: Weight must not affect the platform's transportability requirement. The logger must run on Platform power provided on the CanBus connector (9-32VDC). A full PSPEC will be published in the RFP. Cost is not as important as performance.

Does the data logger need to support all Transmit methods of interest or a subset. FOLLOW ON: if a subset, what is the order of priority and what is critical.

Answer: Manual and WI-FI offload must be supported. Any additional capabilities would be evaluated for their value to the government.

Is the data logger expected to transmit its data (either wired or wireless) in situ, or will it be removed from the platform, data dump, and then reinstalled in its appropriate location

Answer: It is expected to transmit data while installed.

Where does the Fuel Automated Reporting System (FARS) data come from? existing acquired data in a database that we are to re-use? or are you looking to augment data capture thru new sensors, techniques, capabilities, etc. as part of this solicitation. FOLLOW UP: if so, are you envisioning deploying additional sensing and measurement capability for FARS to bring it more in line with this CBM+ program

Answer: FARS hardware is managed by a different Product Management Office. Our responsibility for FARS is the data pipeline, analytics, and visualizations. Data will be transmitted from their hardware to our pipeline. FARS will be a separate and distinct Dashboard

Does the Government find they have sufficient sensor and failure mode coverage with current CANBUS sensors, or will the RFP include a requirement for additional specific sensors or sensor types?

Answer: Adding additional sensors to the platform would be the Platform Owners responsibility based on their RCM analysis

will these slides be shared later?

Answer: Yes. They are now attached to the invite.

Seeing the list of platforms, is the CBM+ team working with any other sensorized weapon systems that are priority to the MAGTF/GCE?

Answer: ACV and ULTVs that have the LMADIS system installed

What percent of initial target platforms (MTVR, JLTV, LVSR, etc.) currently have onboard data loggers installed

Answer: Approximately 10%

Will commercial experience be considered for past performance?

Answer: Yes, if the experience relates to components (Engines, Transmissions, ETC.) that are also used in military vehicles.

Will an attendees list be shared? For the purposes of teaming/collaboration

Answer: Yes

You mentioned already having data ingestion, data processing, analytics, and visualization tools. Can you elaborate on what software gaps there are and what non-proprietary software you may be looking for?

Answer: The biggest gap is actual predictive part failures. We can get to the system, but we need to get to the parts level. The MCEN transitions are data and we use DataBricks and Qlik. Models should be compatible with DataBricks.

What level of clearance are you expecting? Secret ?

Answer: Not everyone working on the contract will need a Secret Clearance just those who need it. Everything we are doing right now is CUI, but an eventual move to the Secret level anticipated.

Will the written Q&A document also be shared?

Answer: Yes

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Will OEM ICDs for CANBus info be provided during proposal or at time of award?

Answer: Providing OEM ICDs for the CANBus prior to proposal. We will have to make a legal determination. But it will likely be after award.

is the primary dashboard in Qlik, or Did Crane build out the additional one, previously discussed, in the Azure env?

Answer: The primary dashboard (Enterprise level) is built in Qlik. The Azure dashboard (Maintainer Dashboard) is built using REACT.

Can a list of on vehicle systems connected to CAN be provided?

Answer: Probably. We will have to get this from platform owners. We will attempt to provide this with the RFP, but we are not promising that we can accomplish that.

For hardware delivery, the documents currently indicate 3000 units per ordering period. How long is an ordering period?

Answer: Quantities will be dependent on funding and time. 3000 units over the three year period is expected.

What capabilities is the current ADVANA/QLIK platform not providing that needs to be provided by the CBM+ solution ? Is MARCORSYSCOM open to modern innovative Analytics Platforms that enable data-driven decision making?

Answer: We are not aware of any gap in capability provided by Advana. MARCORSYSCOM is not seeking a proprietary solution/analytics platform

Will a list of anticipated end user roles be provided?

Answer: Yes

Which OEM vendor has provided the USMC with their current installment base of data loggers? Are you looking for equivalent loggers in the future.

Answer: Trident. Given the speed at which this technology changes there will be a future time when more robust hardware is available. We will be looking for new technology for a possible tech refresh should it provide additional benefit.

Are the existing data loggers capturing and offloading the full set of CAN bus messages available on the platform, or only capturing a subset of interest?

Answer: We are recording everything available on the CANBus. They can be configured to provide a subset.

Are we able to have access to targeted ground system ICDs, for the preparation of the proposal?

Answer: No. ICDs will be available for the performance of work.

Is any current or future contractor managed maintenance being performed by FSRs or is all Maintenance for platforms in CBM+ program done by Marines.

Answer: There is a mix across the USMC. CBM+ FSRs would only be concentrated on our hardware and software. Some platforms do have FSRs that assist Marines, but do not maintain the platform for the USMC.

Does everything that comes off of the CANBus get stored in a single S3 bucket?

Answer:

- a. All recorded data files are sent to the same S3 bucket at the motor pool local unit server (V1) or Azure landing zone (V2).**
- b. Data files are moved to new S3 buckets as they move through the processing pipeline.**
- c. Files are separated by folders (S3 prefixes) labeled with data logger Device ID (pre-conversion) or vehicle Serial Number (post-conversion).**

What file format do you current store the data in?

Answer:

- a. CBM+ V1:**
 - i. Files offloaded from data logger in MFC (compressed).**
 - ii. MFC files decompressed to MF4.**
 - iii. MF4 converted to Parquet.**
- b. CBM+ V2:**
 - iv. Files offloaded from data logger in MFM (encrypted and compressed).**
 - v. MFM files decrypted and decompressed to MF4.**
 - vi. MF4 converted to Parquet.**

What is the approx daily data volume that is received?

Answer:

- a. Approximately 21.6 GB/month for MF4 data files transmitted from all local unit servers to PSU SIL.**
- b. Approximately 0.7 GB/day MF4 data volume across all enrolled motor pools.**

Is supply chain data/ repair parts in stock available for CBM+ analysis / models ?

Answer: Supply stock data is available in GCSS and we can get parts data from DLA. We have not integrated this into the current system.

What are the requirements on the data science / AI side behind the dashboards or applications? Does it include building AI/sensor models and iterating on efficacy testing with maintainers/engineers?

Answer: We require the capability to predict failure probability and remaining useful life, conduct root-cause analysis, and generate predictive advisories that inform prescriptive maintenance recommendations.

This will include building AI models to accomplish these requirements. The development process should involve iterating on model efficacy with feedback from our maintainers and engineers. We view this collaborative loop as essential for making the CBM+ capability practical, trustworthy, and ultimately valuable to the end user.

Is there a requirement to help drive RCB, inventory planning, or other actioning of insights generated?

Answer: Yes. We want to be part of Global Logistics Awareness and planning. We want to be able to have Marines provide us a list of equipment they are taking and we can give them a list of parts to take.

In regards to Caterpillar Heavy Equipment /MHE, are those platforms included in the 1000 currently fielded CBM+ vehicles/equipment?

Answer: Yes

Will the contractor be allowed to decode "ALL" data read on the CAN bus?

Answer: Some of this data is OEM proprietary. So even if you can decode it we might not be able to display it on a dashboard

If you are already able to collect and transmit the data off of the vehicle why are you looking for a new data logging hardware solution? is there some expanded capability that is being sought? is it possible that you will remain with the current hardware solution being used?

Answer: The current hardware solution does what we need it to do. We are aware that there are other solutions out there that have expanded capability. Some platforms have more than two CanBus. We are looking to see if there are some options with out there that we can evaluate.

Did you say that intent is Data Loggers for this effort will have no proprietary software on them?

Answer: The function of the data logger is to collect store and transmit the data. If there is proprietary software that is part of that process it's not a showstopper if it can provide the correct file format for ingestion into the data pipeline.

Does the proposed data logger need to capture ALL CAN bus data or is a CBM+ centric subset desired?

Answer: All data is collected.

If MARCORSYSCOM eliminates Software as Service or Software As a Subscription solutions, what type of Analytics Platforms are left to provide the capabilities? Open Source Solutions? Perpetual software licenses? GOTS?

Answer: Right now we can perform these analytics using tools provided by the Advana Jupiter environment. We have not needed a SaaS platform to get to where we are today.

Have CBM+ efforts to date produced a meaningful volume of coincident failure records and CANbus data logs (i.e. where the operation of the failed part was observed via logged data for its entire useful life)?

Answer: No

Does the current Trident data logger capture data from the Caterpillar Diagnostic Link (CDL) found on equipment such as 120M grader, D6 dozer, 320D grader, etc.

Answer: If this equipment has a SAE Standard CANBus current data loggers should be able to collect the data. We may have to adapt a cable.

You mentioned Trident ... can you clarify if A2DL or PMA hardware?

Answer: Current hardware is the A2DL.

Does the current data logger capture J1939 as well as J1708/J1587 and OBD-I such as found on the HMMWV transmission. Is the HMMWV a target CBM+ platform?

Answer: Yes. But the HMMWV is not a CBM+ targeted platform.

You mentioned 3 years of historical data available, is more failure data available from a different source?

Answer: Yes. GCSS has all historical data. We have 8 years of historical data available to us in Advana

Does GCSS data go down to the serial number of the failed component or more generalized by part number?

Answer: No. GCSS will only provide part number day.

where is the failure data coming from?

Answer: GCSS-MC

Are you open to using proprietary software platform tools that are available currently in Jupiter Advana to enhance and scale Databricks models?

Answer: No

We would like to understand how you intend to acquire the solution if “The desire is not to license any portion of the solution” along with separate statement indicating that you do not want a “software as a subscription” solution. Also, there appears to be a desire to leverage analytic software available in ADVANA, but the software in ADVANA is made available through a software license or subscription which would be in conflict with the statements above.

Answer: The Government intends to acquire the solution through services whereby data scientist, engineers, etc. write analytic models and data ingestion pipelines in the cloud environment using available tools. The software available in Advana (DataBricks and Qlik) are provided at no additional cost to us through the Navy.